

Instrumentation and Control

CSE 325

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Transducers

Transduction:

The process of converting one form of energy into another.

Transducer:

The device that performs this conversion.

Why Use Electrical Transducers?

Most real-world parameters are non-electrical (e.g., pressure, temperature, displacement, force).

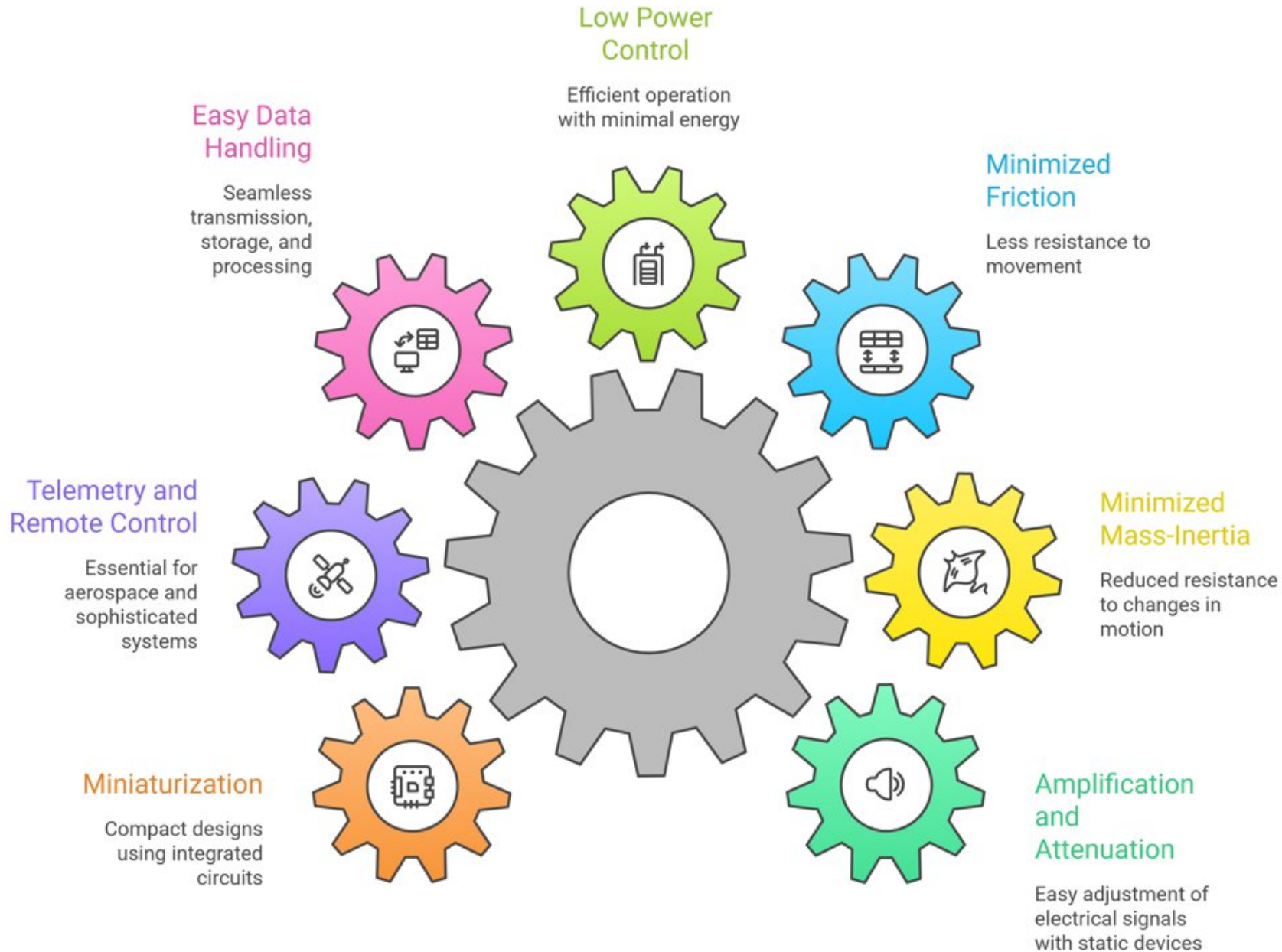
Electrical methods of measurement offer major advantages:

- Better sensitivity, accuracy, and speed
- Easier signal conditioning, amplification, and transmission

Electrical Transducer Definition

“A transducer is a device which, when actuated by energy in one form, produces an electrical output energy.”

Advantages of Electrical Transducer



Advantages of Electrical Transducers

Classification of Transducers

A **classification** can be done **based on the electrical circuit parameter** that is meant to be **changed due to the presence of the physical parameter** to be measured:

1. Resistance
2. Capacitance
3. Inductance
4. Voltage and Current

Primary transducers

Transducers that **sense a physical phenomenon** at the **first stage of detection**; e.g. thermocouple

Secondary transducers

The physical phenomenon is **first sensed by a non-electrical transducer**. It's **output is then converted into an electrical signal** by an electrical transducer. This second transducer is called secondary transducer.

For example: in one type of pressure meter, **Stage 01: Bourdon Tube** € **Input:** Pressure (non-electrical) and **Output:** Mechanical displacement

Stage 02: Secondary Transducer – Electrical Sensing € **Input:** Mechanical displacement from Bourdon tube and **Output:** Electrical signal (voltage, current, or digital)

Resistive Transducers

Transducer that **employs change in resistance of a sensing element** is called resistive transducer.

The well-known equation of resistance of a resistor element:

$$R = \frac{\rho L}{A}$$

Design basis: vary any combination of the **3 quantities**

For example: **temperature can change ρ ; strain can change L and A, altogether**

Applications:

Temperature, strain, displacement, force, pressure etc.

Resistance Temperature Detector (RTD) Or Resistance Thermometer

- An RTD is a **passive temperature sensor** that works on the principle that the **electrical resistance of metals changes predictably with temperature**.
- Most metals exhibit a **positive temperature coefficient (PTC)** – resistance increases as temperature increases.

Resistance Temperature Detector (RTD)

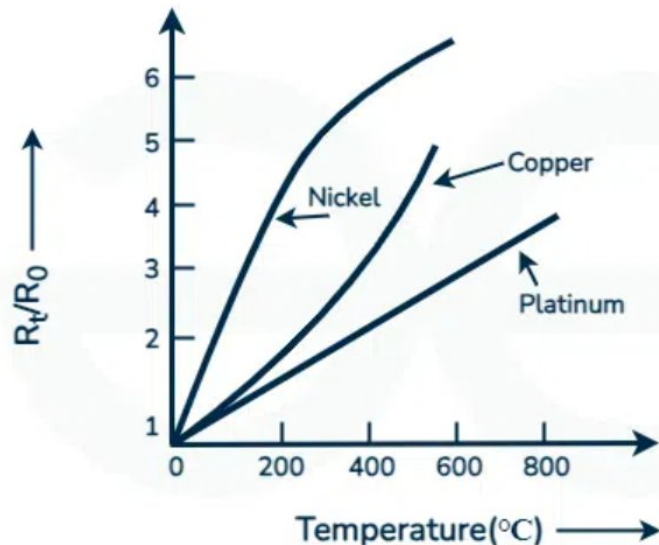
Variation of resistance R with temperature T ($^{\circ}K$) can be represented by the following well-known relationship:

$$R = R_0(1 + \alpha_1 T + \alpha_2 T^2 + \dots + \alpha_n T^n + \dots)$$

R_0 = resistance at temperature $T = 0$

and, $\alpha_1, \alpha_2, \alpha_3$ are constants

Resistance-Temperature characteristics of commonly used RTD metals are plotted (not-exact) here:



The curve shows that the curves are **nearly linear**, or at least can be approximated **as linear for short temperature spans**.

Resistance Temperature Detector (RTD)

Table: Metals used for RTD

Metal	Resistance Temperature Co-efficient (/°C)	Temperature range (°C)	Melting point (°C)
Platinum	0.39	(-260) – (110)	1773
Copper	0.39	(0) – (180)	1083
Nickel	0.62	(-220) – (300)	1435
Tungsten	0.45	(-200) – (1000)	3370

Desirable characteristics of an RTD material:

- Change in resistance per unit change in temperature should be as large as possible.
- High resistivity, so that a minimum volume of material is used for the construction of RTD
- The R versus T curve should be continuous, stable and repeatable.

Resistance Temperature Detector (RTD)

Material	Use in RTD	Key Characteristics	Application Notes
Platinum (Pt)	Widely used (PRTD)	High temp stability, noble metal, resistant to contamination, high accuracy	Best for precision measurement, industrial standard
Gold (Au)	Rarely used	Low resistivity, expensive	Not practical for RTDs
Silver (Ag)	Rarely used	Low resistivity, expensive	Not practical for RTDs
Tungsten (W)	Limited use	High resistivity, extremely brittle, difficult to work	Reserved for very high temperature applications
Copper (Cu)	Used as alternative	Low resistivity, good linearity, low cost	Requires longer element length; economical choice
Nickel & Nickel Alloys	Commonly used	Moderate resistivity, good sensitivity	Common in mid-range temperature applications

Equation: Linear Approximation

For some metals, **linear relationships** can exist over a specified span of the curve

$$R_{\theta} = R_{\theta_0} (1 + \alpha_{\theta_0} \Delta\theta) \text{ for } \theta_1 \leq \theta \leq \theta_2$$

- Two unknowns are here, so two points in the line are necessary to solve them.

A **quadratic approximation of the Resistance-Temperature** curve can be more accurate over a limited range of temperature. It has both **1st** and **2nd** order temperature terms.

$$R_{\theta} = R_{\theta_0} \left[1 + \alpha_1 \Delta\theta + \alpha_2 (\Delta\theta)^2 \right] \text{ for } \theta_1 \leq \theta \leq \theta_2$$

Where, α_1 = Linear fractional change in resistance, / °C

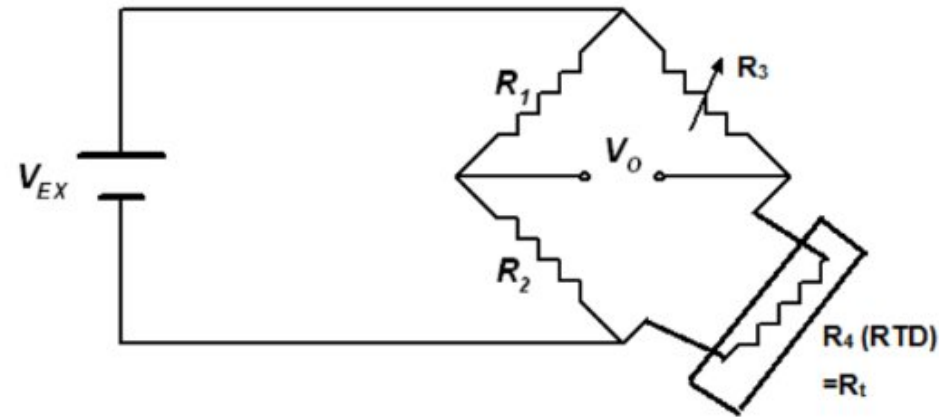
α_2 = quadratic fractional change in resistance, /(°C)²

- Values of the three unknowns (α_1 , α_2 and R_{θ_0}) can be found from tables or graphs by using three points.

Measurement of RTD Resistance

Measurement Bridges are used for this purpose.

Wheatstone Bridge:



RTD is connected as one of the arms of the bridge.

So,

$$R_t = R_3 \left(\frac{R_2}{R_1} \right)$$

Potentiometer R_3 can be calibrated in terms of temperature directly.

Applications

Why RTDs Are Used

- RTDs are among the **most accurate and precise temperature sensors**.
- Many industrial processes operate **at high temperatures where extreme accuracy is not essential** (e.g., $\pm 2\text{ }^{\circ}\text{C}$ error in a $250\text{ }^{\circ}\text{C}$ process is acceptable).
- Some applications, however, **require very high precision — RTDs are preferred** in these cases.

RTD Applications

1. Bioreactors

Requirement: Precise temperature control for growing biological cells.

Why RTD?: Small temperature deviations can kill cells or cause defective growth.

RTD provides: Repeated accuracy and stability for exact temperature regulation.

2. Low-Temperature Processes

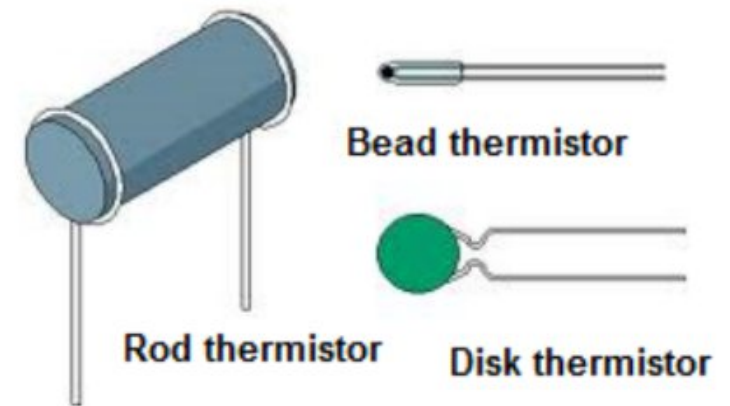
Examples: Pharmaceutical storage, Chemical synthesis, Climate chambers

Requirement: Detecting small temperature changes reliably.

Why RTD? High sensitivity and repeatability at lower temperature ranges.

Thermistors

- Contraction of the term, “thermal resistors”
- Generally **made of semiconductor** materials
- Have **negative temperature co-efficients**, i.e. **resistance decreases with increase in temp** (Due to production of higher number of electron-hole pair generation as temperature increases)
- **Sensitivity is high**, so suitable for very small changes in temperature. This makes them useful for precision temperature measurement applications.
- Relatively **low cost and small size**.
- Measurement range: approximately, $(-60) - (15) ^\circ\text{C}$



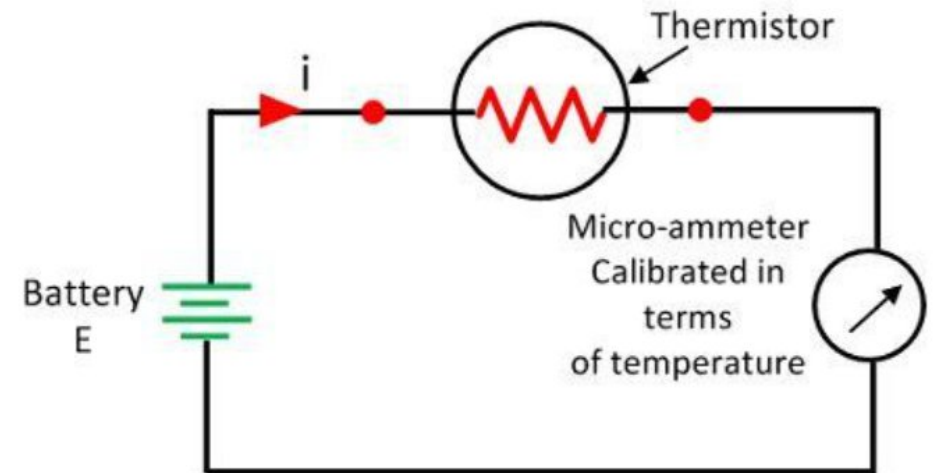
Measurement and Control of Temperature: Thermistors

High Sensitivity

- Provides a **large change in resistance for a small change in temperature**.
- Much higher sensitivity compared to RTDs and thermocouples.
- Ideal for detecting **small temperature variations**.

Simple Circuit:

- Thermistor resistance changes with temperature → current changes.
- **Micro-ammeter can be directly calibrated** in temperature units ($^{\circ}\text{C}$).
- Simple, low-cost, but less precise.



Series Circuit For Measurement of Thermistor

Circuit Globe

Thermistors : Applications

Temperature compensation:

- Thermistors have negative temperature co-efficients, opposite to that of electrical resistors. So, they are widely used to compensate for the effects of temperature.
- They are so connected in the circuit that the total circuit resistance is constant over a wide range of temperatures

Other Applications:

Measurement of flow, providing time delay etc.

Thermistors are versatile not only for sensing but also for actively compensating temperature effects in electronic systems.

